Code: 20A05702b **R20**

B.Tech IV Year I Semester (R20) Supplementary Examinations April/May 2025

CRYPTOGRAPHY & NETWORK SECURITY

(Common to CSE & CSE (Data Science))

Time: 3 hours Max. Marks: 70

PART – A

(Compulsory Question)

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1		Answer the following: (10 X 02 = 20 Marks)	
	(a)	Define Availability. Give an example.	2M
	(b)	Explain briefly the security services.	2M
	(c)	Define modular arithmetic.	2M
	(d)	Define digital signature.	2M
	(e)	Define message digest.	2M
	(f)	Explain briefly the Message Authentication Code (MAC).	2M
	(g)	Explain briefly the user authentication.	2M
	(h)	Tunnel Mode and Transport Mode Functionality of IPSec.	2M
	(i)	Explain briefly the web security threats.	2M
	(j)	Explain briefly the Secure socket layer	2M

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

2		With an example explain shift row transformation technique.	10M
		OR	
3	(a)	Explain briefly the OSI security Services.	5M
	(b)	Explain in detail the transportation technique to encrypt the messages.	5M
4		With neat diagram the public key cryptosystems.	10M
		OR	
5		Explain in detail the Elliptic curve cryptography.	10M
6		Explain in detail the digital signature algorithm.	10M
		OR	
7		With neat diagram explain in detail the PKI architectural model.	10M
8	(a)	With neat diagram explain briefly the Kerberos.	5M
	(b)	With header format explain in detail the IKE messages.	5M
		OR	
9	(a)	With neat diagram the IPSec architecture.	7M
	(b)	Explain briefly the two significant differences between S/MIME and OpenPGP.	3M
10	(a)	Explain in detail the firewall characteristics and policies.	5M
	(b)	With neat diagram the TLS architecture.	5M
		OR	
11	(a)	Explain two types of port forwarding in SSH.	5M
	(b)	Explain briefly the firewall locations and configuration.	5M

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Code: 20A05702b **R20**

B.Tech IV Year I Semester (R20) Regular & Supplementary Examinations December/January 2024/2025

CRYPTOGRAPHY & NETWORK SECURITY

(Common to CSE (DS) & CSE)

Time: 3 hours Max. Marks: 70

PART – A

(Compulsory Question)

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1		Answer the following: (10 X 02 = 20 Marks)	
	(a)	Explain briefly the brute force attack.	2M
	(b)	List all the main parameters of AES.	2M
	(c)	Define asymmetric encryption.	2M
	(d)	Mention the applications of public cryptosystem.	2M
	(e)	What are the key components of NIST digital signature algorithm?	2M
	(f)	Define hash function.	2M
	(g)	Briefly explain the benefits of IPsec.	2M
	(h)	Define the functions of Kerberos.	2M
	(i)	Explain the TLS attacks.	2M
	(j)	Explain briefly the Handshake protocol.	2M

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

2	(a)	With neat diagram explain the model for network security.	4M
	(b)	With neat diagram the DES Encryption.	6M
		OR	
3	(a)	With neat diagram explain the model of symmetric cryptosystem.	6M
	(b)	Define threat and attack? Give an example for each.	4M
4		Explain the Chinese remainder theorem.	10M
		OR	
5		With neat diagram the man-in-the-middle attack.	10M
6		Explain in detail message authentication functions.	10M
		OR	
7		With neat diagram explain in detail the x.509 certificate.	10M
8	(a)	With neat diagram explain briefly the authentication architectural model.	5M
	(b)	With neat function flow diagram explain in detail the S/MIME.	5M
		OR	
9	(a)	With a packet format explain in detail the ESP.	5M
	(b)	Explain in detail types of IP sec documents.	5M
10	(a)	Explain in detail the HTTPS.	5M
	(b)	With neat diagram the SSH Protocol stack.	5M
		OR	
11		Explain in detail the firewall types.	10M

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Code: 20A05702b **R20**

B.Tech IV Year I Semester (R20) Supplementary Examinations July/August 2024

CRYPTOGRAPHY & NETWORK SECURITY

(Computer Science and Engineering)

Time: 3 hours Max. Marks: 70

PART – A

(Compulsory Question)

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1		Answer the following: (10 X 02 = 20 Marks)	
	(a)	List the types of substitution techniques.	2M
	(b)	Define AES.	2M
	(c)	Write a note on Discrete Logarithms.	2M
	(d)	What is Fermat's Theorem?	2M
	(e)	Define hash function.	2M
	(f)	What is Cipher based Message Authentication Code?	2M
	(g)	How IP Security will works?	2M
	(h)	Define MIME.	2M
	(i)	List out the types of firewalls.	2M
	(j)	What is the use of transport layer security?	2M

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

2	(a)	What are security attacks in cryptography?	5M
	(b)	Explain the symmetric cipher model.	5M
		OR	
3		Define network security. With a neat sketch explain the network security model and its components.	10M
4		What is public key cryptography? Explain public key cryptography algorithms with an example?	10M
		OR	
5	(a)	How does Euclidean Algorithm works?	5M
	(b)	Explain the Multiplicative Inverse in GF (p) with an example?	5M
6		What is digital signature? Explain the NIST Digital signature algorithms?	10M
		OR	
7	(a)	List the application of cryptographic hash functions.	5M
	(b)	Write a short note on public key infrastructure.	5M
8	(a)	Distinguish PGP vs S/MIME.	5M
	(b)	Explain IP Security?	5M
		OR	
9		What is Encapsulating Security Payload (ESP) and how does it works? Explain the modes and components in ESP?	10M
10		With a neat sketch explain the firewall location and configuration.	10M
		OR	
11		Brief note on (i) TLS, (ii) SSH, (iii) HTTPS.	10M

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Code: 20A05702b

R20

B.Tech IV Year I Semester (R20) Regular Examinations December/January 2024

CRYPTOGRAPHY & NETWORK SECURITY

(Common to CSE (DS) and CSE)

Time: 3 hours | Max. Marks: 70

PART – A

(Compulsory Question)

*****

1		Answer the following: (10 X 02 = 20 Marks)	
	(a)	Define Steganography.	2M
	(b)	What is privacy?	2M
	(c)	Explain Euler's theorem.	2M
	(d)	Define Elliptic Curve Cryptography.	2M
	(e)	How digital signature is important in Cryptography?	2M
	(f)	List some applications of cryptographic hash function.	2M
	(g)	Define Kerberos.	2M
	(h)	How to encapsulate security payload?	2M
	(i)	Write a note on firewall.	2M
	(j)	Define SSH Protocol.	2M

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

2		With a neat sketch Explain the OSI Security Architecture.	10M
		OR	
3	(a)	What is AES and how does it works? Why the AES algorithm is necessary?	5M
	(b)	Explain the data encryption standard.	5M
4		What is Diffie - Hellman key exchange and how does it works? Consider Diffie - Hellman key exchange between two parties A and B. Let A sends J = Nx1 (mod p) and B sends K = Nx2 (mod p) respectively. Let N = 5, x_1 = 3, x_2 = 6 and p = 23. How to calculate and find the shared key between two parties.	10M
		OR	
5	(a)	Demonstrate the working process of RSA Algorithm.	6M
	(b)	Explain Chinese remainder theorem.	4M
6	(a)	When to use the Secure hash algorithm in cryptography?	5M
	(b)	Write a brief note on X.509 certificate.	5M
		OR	
7	(a)	Distinguish HMAC and CMAC.	5M
	(b)	Explain message authentication function.	5M
8		How does the electronic mails will be secure? How to choose the type of electronic mail security with an example?	10M
		OR	
9	(a)	What is Internet key exchange?	5M
	(b)	List the principles of remote user authentication.	5M
10		What is firewall? Explain the types of firewalls with an example. How the firewall will works?	10M
		OR	
11		Brief note on: (i) HTTPS, (ii) TLS.	10M

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